

Convergence and Divergence Tests for Series

Test	When to Use	Conclusions
Divergence Test	For any series $\sum_{n=0}^{\infty} a_n$	Diverges if $\lim_{n \rightarrow \infty} a_n \neq 0$.
Integral Test	$\sum_{n=0}^{\infty} a_n$ with $a_n \geq 0$ and a_n decreasing	$\int_1^{\infty} f(x) dx$ and $\sum_{n=0}^{\infty} a_n$ both converge/diverge where $f(n) = a_n$.
Comparison Test	$\sum_{n=0}^{\infty} a_n$ and $\sum_{n=0}^{\infty} b_n$ if $0 \leq a_n \leq b_n$.	$\sum_{n=0}^{\infty} b_n$ converges $\implies \sum_{n=0}^{\infty} a_n$ converges. $\sum_{n=0}^{\infty} a_n$ diverges $\implies \sum_{n=0}^{\infty} b_n$ diverges.
Limiting Comparison Test	$\sum_{n=0}^{\infty} a_n$, ($a_n > 0$). Choose $\sum_{n=0}^{\infty} b_n$, ($b_n > 0$). If $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = L$ with $0 < L < \infty$	$\sum_{n=0}^{\infty} a_n$ and $\sum_{n=0}^{\infty} b_n$ both converge/diverge.
	if $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = 0$	$\sum_{n=0}^{\infty} b_n$ converges $\implies \sum_{n=0}^{\infty} a_n$ converges.
	if $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \infty$	$\sum_{n=0}^{\infty} b_n$ diverges $\implies \sum_{n=0}^{\infty} a_n$ diverges.
Convergent Test for Alternating Series	$\sum_{n=0}^{\infty} (-1)^n a_n$ ($a_n > 0$)	converges if $\lim_{n \rightarrow \infty} a_n = 0$ and a_n is decreasing.
Absolute Convergence	for any series $\sum_{n=0}^{\infty} a_n$	If $\sum_{n=0}^{\infty} a_n $ converges, then $\sum_{n=0}^{\infty} a_n$ converges. (definition of absolutely convergent series.)
Conditional Convergence	for any series $\sum_{n=0}^{\infty} a_n$	if $\sum_{n=0}^{\infty} a_n $ diverges but $\sum_{n=0}^{\infty} a_n$ converges, $\sum_{n=0}^{\infty} a_n$ conditionally converges.
Ratio Test	for any series $\sum_{n=0}^{\infty} a_n$, calculate $r = \lim_{n \rightarrow \infty} \left \frac{a_{n+1}}{a_n} \right $	There are three cases: if $r < 1$, then $\sum_{n=0}^{\infty} a_n $ converges; if $r > 1$, then $\sum_{n=0}^{\infty} a_n $ diverges; if $r = 1$, no conclusion can be made.
Root Test	for any series $\sum_{n=0}^{\infty} a_n$, calculate $\rho = \lim_{n \rightarrow \infty} \sqrt[n]{ a_n }$	There are three cases: if $\rho < 1$, then $\sum_{n=0}^{\infty} a_n $ converges; if $\rho > 1$, then $\sum_{n=0}^{\infty} a_n $ diverges; if $\rho = 1$, no conclusion can be made.

Important Series to Remember

Series	How They Look	Conclusions
Geometric Series	$\sum_{n=0}^{\infty} ar^n$	Converges to $\frac{a}{1-r}$ if $ r < 1$ and diverges if $ r \geq 1$.
p -Series	$\sum_{n=1}^{\infty} \frac{1}{n^p}$	Converges if $p > 1$ and diverges if $p \leq 1$.

Power Series to Memorize

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \dots$$

$$\frac{1}{1+x} = \sum_{n=0}^{\infty} (-1)^n x^n = 1 - x + x^2 - x^3 + x^4 - x^5 + \dots$$

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

$$\sin(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

$$\cos(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$$

$$\sinh(x) = \sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!} = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \frac{x^7}{7!} + \dots$$

$$\cosh(x) = \sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!} = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \frac{x^6}{6!} + \dots$$

$$(1+x)^k = \sum_{n=0}^{\infty} \binom{k}{n} x^n = 1 + kx + \frac{k(k-1)}{2!} x^2 + \frac{k(k-1)(k-2)}{3!} x^3 + \dots$$

The first two and last one have radius of convergence 1, while the others have radius of convergence ∞ .

Also, here are some power series which are handy to know, but can be derived on the fly, if necessary, by integrating:

$$\tan^{-1}(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1} = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots$$

$$\ln(1+x) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^n}{n} = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$$

Both have radius of convergence 1.